

Hey, everybody. Welcome to your chapter 10 review. In this episode, we're going to cover supplementation. Supplementation is a huge industry. However, the concept of performance and wellness-enhancing foods or substances is not new. Supplements have been around for thousands of years. Ayurvedic medicine has been practiced in India dating back to 6,000 BC. And in 300 BC, Roman gladiators concocted their own type of recovery shake made of plant ash. Sounds delicious, right?

Supplements as we think of them today originated in the 1940s with a simple multivitamin product. Since then, and as of 2018, the dietary supplement industry has grown to generate \$115 billion of revenue worldwide, most of which continue to be vitamin and mineral sales. By 2025, sales are expected to swell to nearly \$200 billion.

The variety of supplements available has broadened to include a great deal of fitness-related products. You'll find supplementation is often viewed as part of the fitness package. In the gym setting, fitness professionals may be asked to sell supplements. If you don't agree with that, you may need to find a different place to work that doesn't conflict with your views. You may also find opportunities as an influencer for supplement companies on social media or other online platforms.

You're going to be approached by clients, social media connections, and that well-meaning relative at the holiday gathering asking for your expert opinion. So it's important for you to have a general understanding of the uses, regulations, and capabilities of the options out there. With that being said, it's also important to note that while supplement use might seem commonplace in the fitness community, fitness professionals must know and stay within their scope of practice, understanding their responsibilities and liabilities as a CPT.

You should be a filter for your clients, educating them on the basics of supplements while emphasizing that supplements are just another component to an overall exercise program and won't get them to their goal on their own. There are no quick fixes. And we never prescribe supplements to treat a medical condition or disease. By the end of this chapter, you should be able to define supplementation and the dietary need for taking them, summarize the supplements used for optimal health and body composition, as well as explain the role of supplementation within the personal training profession.

That's your high-level overview. Now let's zoom in and see what else there is to uncover. The technical definition of dietary supplements are products intended to supplement the diet that contains one or more vitamin, mineral, amino acid, herb, or other botanical. They are used by individuals to supplement the diet by increasing the total dietary intake or a concentrate of a specific nutrient. They are used for a wide range of reasons, and some are more essential for human nutrition than others.

They can lead to toxicity if overconsumed. So it's vital to consider your client's overall health status when discussing supplements and refer them to a physician when needed. In the US, dietary supplements are regulated by the FDA according to the Dietary Supplement Health and Education Act. However, supplements don't require review or approval prior to being marketed and sold. So you'll need to understand the required components of the dietary supplement label, including the active ingredients, other ingredients, pertinent warnings, total contents, usage instructions, and recommended serving size.

While regulations for dietary supplements do exist, supplements can be brought to market in most countries without prior government approval. This results in potential safety issues, like new and untested ingredients, containing undisclosed ingredients, or containing illegal ingredients, like banned stimulants or anabolic steroids. Not all dietary supplements should be trusted unconditionally. Make sure you're reading labels, reviewing published literature, and choosing only those with third-party verification, like NSF Certified for Sport certification and other evidence to support their safety and effectiveness.

It's also a good idea to identify the effective dosage of common dietary supplements like protein, creatine, and caffeine to readily identify dietary supplements on the market with adequate dosages. Don't panic. You don't need to remember exact daily values or tolerable upper limits. But you should be wary of DVs listed in multiples of 100, as these may exceed the tolerable upper limit of a supplement, resulting in adverse effects or toxicity.

Dietary supplements can be classified into two broad categories, health supplements and performance supplements. Health supplements include vitamins, minerals, and omega-3 products, to name a few. Dietary supplements used specifically for performance are classified as ergogenic aids. Many supplements claim to improve fat loss, strength, or muscle gain, but few are substantiated. You must be able to guide your clients toward safe supplements that may yield results consistent with their goals and away from unsafe or counterproductive supplements.

Most of the dietary supplements currently sold are multivitamins, which contain a variety of vitamins and minerals. Individual vitamins and minerals are available as well for those who have identified specific needs. This category of dietary supplements is often used to ensure adequate vitamin and mineral intake, as micronutrient intake is typically inadequate with Western diets as well as more restrictive diets.

Often overlooked, vitamins are available in both biologically active and inactive forms. In some cases, the inactive form may be converted to the active form. This is important because inactive forms are less efficient for augmenting body levels and achieving desired outcomes. You also have both your water- and fat-soluble vitamins, and it's important to understand the details of both categories. For instance, water-soluble vitamins are more transient compared to fat-soluble vitamins, meaning they are less likely to produce toxicity and it is easier to treat acute deficiencies.

Omega-3 fats are one of the more thoroughly researched dietary supplements available. Often supplemented as fish oil, omega-3 supplements can help in preventing cardiovascular disease, reducing inflammatory status, and optimizing neurological development. They are the third most frequently used dietary supplement. And you can definitely anticipate discussing them with your clients.

Protein supplements are convenient to help a client increase their total daily protein intake. So trainers need to understand the different types of proteins and protein needs based on activity levels and fitness goals. As more intense exercise causes more muscle damage, the body will require additional protein to facilitate recovery. Clients who were previously inactive will also have increased protein needs. So be sure this is a topic of conversation with clients who are going from a fairly sedentary lifestyle to getting back in the swing of exercising regularly.

Alternatively, many clients will have goals that center on building muscle size or strength. Protein and amino acids may help facilitate those goals. Creatine is another supplement with substantial evidence for adding mass, strength, and fatigue resistance. Creatine supplementation is a simple way for individuals to maximize their training potential. For those clients that are willing to push their physical limits, it may help them perform additional reps and increase their training volume.

Creatine is a naturally produced compound made from three amino acids. It is naturally produced by the liver and kidneys, but it's primarily stored in skeletal muscle, where it is able to exert its primary function. As a supplement, it is stored in many convenient forms, such as powders and capsules. Creatine is not known to cause toxicity effects, but it also poses no greater benefit when the consumed amount exceeds the recommended dosage.

Of all the dietary supplements, caffeine is the one most likely already being used by your clients. It is found in coffee, tea, energy drinks, pre-workout supplements, energy shots, and gum. Most will use it to combat tiredness and maintain a better level of daily functioning. For some, that may also mean using caffeine as part of their exercise routine. Caffeine increases wakefulness, attention, and focus by chemically blocking receptors that create the sense of feeling tired.

It also causes an adrenaline release, which increases metabolic rate and subsequently leads to more energy by our good pal, ATP, being metabolized. So far as supplements go, those are the big ones. Here's another little supplement offshoot for you. Depending on the type of client you're working with-- let's say it's a collegiate or professional athlete \$there, are banned substances to be considered. Banned substances are not to be taken lightly. This group of ingredients are the most dangerous to use.

The supplement itself may be legal. However, they may not be permitted in sports due to the rules of the sports federation. So just make sure your athletes in question are informed by their governing body, whether that's the NCAA, WADA, or whomever, concerning what substances are prohibited from competition. Again, that's where a third-party verification comes in, like Informed Choice or NSF. Any supplement bearing those verifications are basically free of banned substances and will not jeopardize an athlete's standing with the governing body.

So, in summary, know how to interpret a dietary supplement label to recognize potential trouble spots, such as improper labeling, misleading claims, and possible proprietary blends. Look for third-party verification and scientific evidence demonstrating safety and effectiveness. And know the effective daily dosage of common dietary supplements.

And finally, I'll just reiterate one more time that if there is a case with a client where supplements are needed to treat a medical condition or deficiency, it's best to refer them to the appropriate medical professional, whether it's a doctor or a registered dietitian. There is an ethical and legal side to dealing with supplements as well. Well, hopefully this has been supplemental to your studies on the subject. Stay safe out there. Be well. And great job coming this far in the course.